

Physicochemical Transformations of Mixed Fuels Based on Typical Coals and Wood upon Heating

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Abstract—The thermal decomposition of the mixtures of two types of dispersed coals and shredded wood in a temperature range until the complete pyrolysis of the organic matter of both components at different concentrations was experimentally studied in order to evaluate the prospects of using these fuels as composite fuels for large- and small-scale power plants (combustion in the furnaces of steam and hot-water boilers). It was established that the joint thermal decomposition of a mixture of coal and wood particles leads to a significant change in the pyrolysis temperature range and the release of anthropogenic gases (sulfur and nitrogen oxides) in the case of the high-temperature heating of such a mixture based on lean coal. A similar effect, but on a much smaller scale, was detected for a mixture based on long-flame coal. A hypothesis on the mechanism of sequestering sulfur and nitrogen oxides on the thermal decomposition of a mixture of lean coal and wood particles as a result of the interaction of intermediate gaseous and solid products of coal and wood pyrolysis in a temperature range to 1000°C was formulated. The possibility of using a mixture of crushed coal and wood as fuel for steam and hot water boilers was substantiated. With a certain decrease in the energy characteristics of such fuels compared to homogeneous coals, the environmental and economic characteristics of fuel burning processes were significantly improved. It was shown that a significant synergistic effect of co-combustion of particles of coal and wood is achieved only in certain coals.

Keywords: coal, wood biomass, mixed fuel, thermal decomposition, combustion, elemental composition, thermal analysis, mass-spectrometric analysis

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INTRODUCTION

About a third of the world's electricity is generated at thermal power plants that use coal as the main fuel [1] (in China, more than 70% [2]). Although combustion products with high concentrations of sulfur and nitrogen oxides [3, 4] are emitted into the atmosphere on the combustion of this solid fuel, if we look at the trends of the last decade [5, 6], there is no substitute for it as the main fuel at not only many currently operating but also projected thermal power plants [7–9]. In this regard, in recent years, scientists and engineers in many developed countries (China, India, the United States, Japan, Germany, Denmark, Finland, etc.) have attempted to develop technologies for burning coal in the furnaces of steam and hot-water boilers in combination with other combustible substances and noncombustible components (various mixed fuels based on coal) [9–13]. These fuels include coal–water [9–11] and organic coal–water mixtures [12–15] and the mixtures of coal with biomass [16, 17]. At first glance, these latter are the simplest compositions for burning in furnaces. However, despite numerous

(more than 230) attempts made to develop technologies for the burning of mixed fuels based on different grades of coal and different types of biomass at thermal power plants, these compositions are used as the main fuel only at 16 plants. Table 1 summarizes these thermal power plants [18].

Examples of the use of coal–water and organic coal–water fuels in large-scale power generation are much less in number. This state of affairs is largely due to the insufficient (both experimental and theoretical) development of the combustion and, especially, ignition processes of various mixed fuels. For example, theoretical studies on the ignition of coal–water fuels under conditions corresponding to the furnaces of power plants [10, 11] showed that the processes of heat and mass transfer and physicochemical transformations that occur in a drop of coal–water suspension during an induction period are much more complicated than analogous processes in solid, liquid, or gaseous fuels burned in the furnaces of thermal power plants. Experimental data for the mixtures of dispersed coal with crushed biomass are also insufficient

Table 1. Thermal power plants at which mixed compositions are the main type of fuel [18]

Company	Country	Fuel	Electrical power of the plant, MW
Lowa Electric Light and Power	The United States	Coal and agricultural wastes	45
SEPCO	The United States	Coal and waste wood	54
Stockholm Energy	Sweden	Coal, wood, and olive waste	54
Ames Municipal Electric	The United States	Coal and fuel from domestic waste	75
Saabergwerke AG	Germany	Coal and domestic waste	75
Georgia Power	The United States	Coal and waste wood	100
GPU GENCO	The United States	Coal and waste wood	130
Tennessee Valley Authority	The United States	Coal and waste wood	150
Midkraft Energy	Denmark	Coal and straw	150
Vasthamnsvert CHP	Sweden	Coal and wood	180
Elsam	Denmark	Coal and straw	250
Baoji	China	Coal and straw	300
Uppsala Energy AB	Sweden	Coal, peat, and wood	320
Lakeland Electric and Water	The United States	Coal and fuel from domestic waste	350
VEAG	Germany	Coal and wood	350
EPON	The Netherlands	Coal and waste wood	602

for the development of combustion technologies, and mathematical models [19] that describe a set of physicochemical and heat and mass transfer processes during the ignition and combustion of structurally inhomogeneous media and materials with consideration for possible changes in a large set of significant factors were not yet developed. Therefore, experimental studies of the basic laws of physicochemical transformations in the mixtures of dispersed coal with biomass under conditions of high environmental temperatures are of considerable current interest.

For example, previously performed experiments with such fuel compositions [16, 17] were mainly focused on the practical applicability of these composite fuels based on coal and biomass.

According to published data [16, 17], the main advantage of the joint combustion of coal and biomass consists in a greater or lesser decrease in the yield of anthropogenic gas oxides formed upon their combustion, as compared with homogeneous coal.

The mechanism of this decrease in the emissions of sulfur and nitrogen oxides into the atmosphere was not identified in the well-known studies [16, 17] (chemical or physical processes for their suppression have not been established).

For these reasons, the aim of this work was to experimentally study the joint thermal decomposition of two types of dispersed coals and crushed wood biomass in a temperature range up to the end temperature of the thermal decomposition of the organic matter of coals, to evaluate the decrease of anthropogenic oxide emissions upon the pyrolysis of such mixtures in com-

parison with that of homogeneous coal, and to formulate a hypothesis on the mechanism (a complex of physicochemical processes) of such a reduction of emissions.

EXPERIMENTAL

Long-flame coal from the Listvyazhnoe deposit and lean coal from the Alardinskoe deposit, which are typical of large- and small-scale power engineering, were chosen as basic components for the test composite solid fuels. Shredded pinewood (OOO Dzerzhinskii LPK, Tomsk) was the second component of the test mixed fuels. Pine sawdust is the main type of wood waste in the territory of the Russian Federation and in many other states. This sawmill waste is very cheap and its use as the second component of composite fuels is cost-effective. As a rule, data on the elemental composition of all components are required for analyzing the thermal decomposition processes of solid fuels based on dispersed coal and wood. For this purpose, we carried out experimental studies to determine the technical and elemental composition of the test coals and wood, and Table 2 summarizes the results.

The experimental studies began with the preparation of fuel. The experiments were conducted under laboratory conditions at an ambient temperature of 20°C, a pressure of 101.4 kPa, and a humidity of 75%.

In the preparation of the experiments, the samples of coal were ground in a ball-type drum mill to a finely dispersed state. The mill included a 5-L ceramic

Table 2. Results of the experimental determination of the heat of combustion, moisture content, ash content, and the yield of volatile matter

Fuel	W^a	A^d	V^{daf}	C^{daf}	H^{daf}	N^{daf}	S^{daf}	O^{daf}	Heat of combustion Q , MJ/kg
	wt %			% on a daf basis					
Coal of grade D	5.8	15.8	12.4	74.2	3.6	1.6	0.1	20.5	26.2
Coal of grade T	5.5	18.4	25.6	84.3	6.4	2.9	0.4	6.0	25.7
Pine wood	5.4	0.3	80.3	58.9	6.9	—	—	34.2	21.7

drum, ceramic grinding balls, and specialized rolls. After grinding, sieving (GOST [State Standard] R 51568-99) and preparation of samples with an average particle size of $<80\ \mu\text{m}$ was performed.

Various impurities (bark, knots, and branches) were preliminarily removed from sawmill waste. The composite fuels were prepared using wood of the same origin (waste of one shift sawing). The preparation of composite fuel samples was carried out by sawdust sampling followed by sieving through a $200\text{-}\mu\text{m}$ sieve.

The weighed portions of particles passed through the sieves formed composite fuels with different volume concentrations of wood mixed with coal of grades D and T in component ratios (%) of 10/90, 25/75, and 50/50. The finished mixture was placed in a zinc-coated drum of a Pulverisette 6 planetary mill with spherical grinding bodies with a diameter of 5 mm in a weight ratio of 1 : 1. The mixing process was carried out at a rotation speed of 500 rpm for 7 min [20].

The elemental composition, form, and structure of the mixed fuels were analyzed by energy dispersive X-ray spectroscopy (EDX) using a JSM 6000C scanning electron microscope (SEM) (JEOL, Japan) [20]. The instrument allowed us to examine the surface structure of the mixed fuels with magnification factors from 10 to 60 000.

The JSM 6000C microscope made it possible to scan the sample surface with a maximally focused electron beam with the simultaneous detection of radiation excited by this beam [20]. Low-energy (to 50 eV) secondary electrons, which were generated by sample material upon relaxation and excitation of external electron shells caused by irradiation, were used as this radiation. This allowed us to display the surface relief of the test sample. To improve the quality of particle images, the particles were covered with a thin layer of gold ($0.02\ \mu\text{m}$), and the accelerating voltage was varied in a range from 10 to 15 kV.

The technical characteristics of fuels (caloricity, ash content, moisture content, and yield of volatiles) are traditionally used to substantiate the possibility of their effective use when burning in the furnaces of power-generating equipment. For the quantitative evaluation of these characteristics of the test fuels, we carried out experimental studies in accordance with procedures described in GOST 147-2013 (ISO 1928-

2009), GOST 11022-95, GOST 27314-91 (ISO 589-81), and GOST 6382-2001 [21–24].

The numerical values of the rates of thermal decomposition of composite fuels and the initial and final temperatures of the thermal decomposition make it possible to evaluate the influence of wood in the composite solid fuels based on the coals of grades D and T on the burning characteristics. The experimental studies were performed on a Netzsch STA 449 F3 Jupiter instrument (Netzsch, Germany) [25]. This instrument allowed us to perform analysis in a temperature range of 25–1300°C. The temperature was increased at heating rates varied in a range from 0.1 to 100 or from 0.1 to 10 K/min [25]. The analyzer operated with blowing an inert gas or reaction gas through the reaction chamber. The thermal analysis of each 20-mg sample was performed at a heating rate of 10 K/min (a maximum temperature reached 1100°C) [25].

To determine the rate of thermal decomposition of mixed fuels, we used a procedure reported by Haykiri-Acma and Yaman [26]. The initial temperature of the thermal decomposition was determined in accordance with a published procedure [27, 28]. The index of the onset of thermal decomposition of fuels (D_i) was calculated from the formula [26]

$$D_i = \frac{V_{\max}}{t_p t_e}.$$

Here, $v_{\max} = (d_w/d_t)_{\max}$ is the maximum rate of thermal decomposition, t_p is the time taken to reach a maximum rate of thermal decomposition, and t_e is the time of the onset of thermal decomposition.

The index D_f was used to evaluate the degree of complete thermal decomposition, which is described by the formula [28, 29]

$$D_f = \frac{V_{\max}}{\Delta t_{1/2} t_p t_f},$$

where $\Delta t_{1/2}$ is the half-period at the moment of reaching a maximum rate of thermal decomposition, and t_f is the time of the complete thermal decomposition of fuel.

An analysis of the degree of reactivity of the test mixed fuels made it possible to determine changes in their characteristics upon varying the fractions of components.

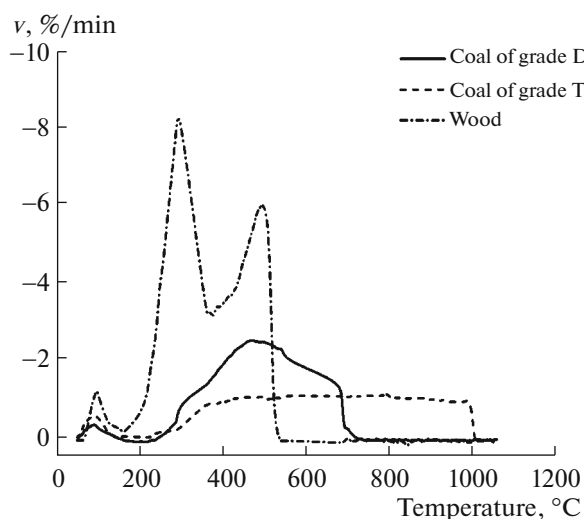


Fig. 1. The temperature dependence of the rate of thermal decomposition of D and T grade coals and wood.

The influence of the fraction of wood in mixed solid fuels with different grades of coal on the yield of volatile substances was estimated by studying the thermal decomposition of these fuels with the use of a QMS 403 C Aeolos quadrupole mass spectrometer [28, 29].

The mass spectrometer is designed for the complex analysis of multicomponent gases, and it allows one to perform the dynamic monitoring of the composition of various mixtures. In the course of analysis, additional equipment was directly connected a Netzsch STA 449 F3 Jupiter thermal analyzer (Netzsch, Germany), and the mass spectra were measured simultaneously with the thermal analysis [25].

The results of the analysis were represented in the form of mass-spectrometric curves corresponding to the main anthropogenic gases (CO_2 , SO_2 , and NO_x) and compared with thermogravimetric and differential thermoanalytical curves. The experimental mass spectra of gases evolved upon the thermal decomposition of the test mixed fuels were compared with the mass spectra of gases formed upon the pyrolysis of homogeneous coals. Based on the results of the comparison, the weight of anthropogenic gases emitted upon the thermal decomposition of composite fuels based on coal and wood was evaluated.

Due to a certain structural inhomogeneity of the two test coals, random errors in the determination of the characteristics of the tested mixed solid fuels were evaluated in small-weight samples (due to possible small deviations in the concentrations of organic and mineral materials in the fuel). For this purpose, all of the experiments were repeated three or four times at fixed values of the main significant factors; the results were averaged and standard deviations were calculated, the relative values of which did not exceed 4–

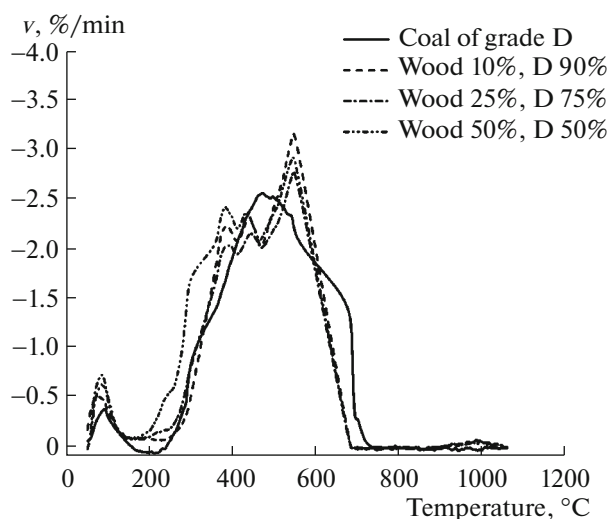


Fig. 2. The temperature dependence of the rate of thermal decomposition of mixed fuels based on D grade coal and wood.

5%. At this stage of experimental data processing, the systematic errors of the instruments used for measuring the main characteristics were also evaluated. These errors did not exceed 1.5% (due to the high level of detection systems in the analytical instruments and measuring equipment used).

Figures 1–3 show the results of the study of the thermal decomposition of the weighed samples of the main components of fuels and composite fuels with coal to wood ratios from 10 to 50%.

The thermograms shown in Fig. 1 make it possible to estimate the initial temperatures and the temperatures of the complete thermal decomposition of the main components used in the formation of composite fuels. The direct thermal decomposition of wood, coal of grade D, and coal of grade T came into play at temperatures from 175, 220, and 240°C, respectively. The thermal decomposition of wood, D grade coal, and T grade coal was complete at temperatures of 550, 750, and 1020°C, respectively. The first peak in the thermograms (Fig. 1) corresponds to the evaporation of water in all fuels. The thermal decomposition of wood exhibited two most pronounced peaks. The decomposition of hemicellulose was complete in a range from 175 to 400°C. The decomposition of cellulose and lignin occurred as the temperature was increased from 400 to 570°C.

Figure 2 presents the results of the thermogravimetric analysis of mixed fuels based on D grade coal and wood. These thermograms allowed us to conclude that an increase in the concentration of wood in the mixed fuel to 50% leads to a decrease in the time of the complete thermal decomposition to 8.3% relative to coal. The fuel decomposition process was complete at a temperature of $\approx 690^\circ\text{C}$. This was due to an increase

in the weight of volatile substances (as compared with coal) upon the heating of composite fuels and lower temperatures of the onset of thermal decomposition of wood.

Figure 3 shows the results of the thermogravimetric analysis of mixed fuels based on T grade coal and wood. By analogy with long-flame coal, the process of the thermal conversion of composite fuels from T grade coal can be tentatively divided into four stages (Fig. 3). At the first stage, moisture is removed; the decomposition of hemicellulose and cellulose occurs at the second stage; the decomposition of lignin and, partially, coal occurs at the third stage; and the decomposition of coal ends at the fourth stage.

Figures 2 and 3 clearly show differences in the thermogravimetric curves of fuels based on long-flame and lean coals. The latter is characterized by a significant decrease in the temperature range in which the pyrolysis of mixed fuel occurs with the fraction of wood. The thermal decomposition of homogeneous T grade coal ended at the temperature $T \approx 1020^\circ\text{C}$. At a wood fraction of 10% in the mixed fuel, the process of the thermal decomposition of mixed fuel was complete at the temperature $T \approx 950^\circ\text{C}$. As the fraction of wood was increased to 25%, the thermal decomposition of composite fuel was complete at a temperature lower than 700°C . In this case, biomass acted as an intensifier of the pyrolysis of lean coal. The above regularities allowed us to conclude that there is a high degree of interaction between the mixed fuels based on T grade coal and woody biomass during their thermal decomposition. It is important that this effect was insignificant in the mixed fuels based on long-flame coal; that is, wood is a selective initiating additive. Accordingly, a thermal analysis similar to that described above should be carried out in the evaluation of the applicability of any coal as a base component in this type of mixed fuels.

Table 3 summarizes the results of the thermogravimetric analysis of mixed fuels based on D and T grade coals and wood. These data make it possible to conclude that an increase in the wood fraction to 50% leads to a decrease in the initial temperature of the thermal decomposition of composite fuel based on D grade coal to 20.9% and to an increase in the maximum reaction rate of thermal decomposition by a factor of 1.7. The same ratio between components in composite fuels based on T grade coal leads to a decrease in the initial temperature of thermal decomposition to 32.5%. The increase in the maximum reaction rate of thermal decomposition of the fuels was as high as 70% (Fig. 3). The period of complete thermal decomposition was decreased to 34.2%. The index of ignition is related to the initial temperature of the thermal decomposition of composite fuels (the lower the initial temperature of the thermal decomposition of fuels, the higher the value of D_i).

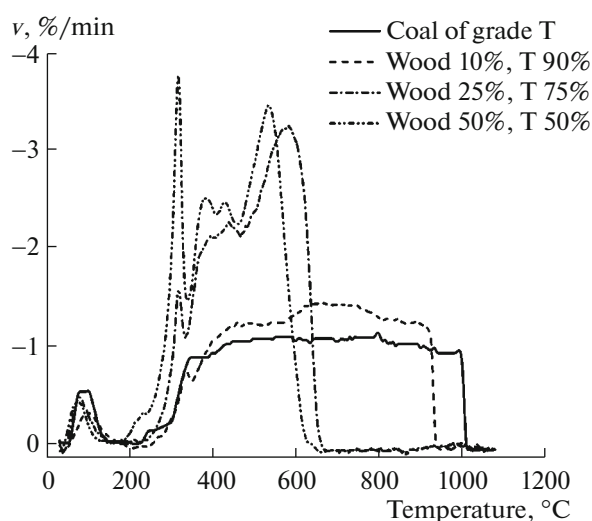


Fig. 3. The temperature dependence of the rate of thermal decomposition of mixed fuels based on T grade coal and wood.

The experimental results allowed us to conclude that an increase in the weight fraction of wood in the mixed fuels with the test coals leads to a decrease in the initial temperatures of thermal decomposition, an increase in the maximum reaction rate of this process, and a shift of the temperature peak corresponding to the maximum rate of pyrolysis of coal to lower temperatures.

Figure 4 shows the results of an experimental analysis of the concentrations of anthropogenic gases in the thermal decomposition products of composite fuels based on D and T grade coals.

Figure 4 indicates that an increase in the fraction of wood to 50% in the mixed fuels led to a decrease in the release of the main anthropogenic oxides NO_x , CO_2 , and SO_2 by 23.8 or 60.2, 17.1 or 85.1, and 38.7 or 89.0% from the fuels based on D and T grade coals, respectively.

The results show the possibility of a significant (to 90%) decrease in the negative impact of these gases from thermal power plants on the environment with the use of mixed fuels based on coal and wood. Based on the results obtained, a hypothesis on the mechanism of reducing the fraction of anthropogenic gases in the combustion products of composite solid fuels was formulated.

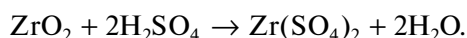
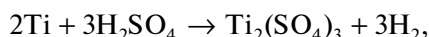
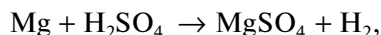
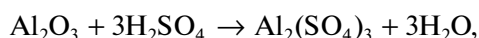
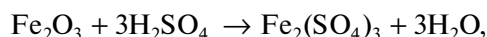
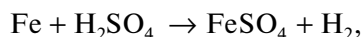
The concentration of carbon oxides decreased due to the fact that wood is a carbon-neutral component. Biomass releases the same amount of carbon as that absorbed during the growth of wood.

The yield of sulfur oxides decreased due to their interaction with water vapor (as a result, sulfuric acid was formed) and the subsequent reactions of metals and their oxides contained in coal with sulfuric acid at high temperatures:

Table 3. Results of the thermogravimetric analysis of the mixed fuels based on D and T grade coals and wood

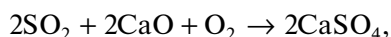
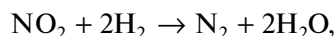
Fuel (wood/coal), %	* $v_{\max}$, %/min	T_p	T_e	T_f	$\Delta T_{1/2}$	D_i , wt %/ T^2	D_f , wt %/ T^3
		°C					
100/0	8.4	304	175	550	267	15.78×10^{-5}	18.82×10^{-8}
0/100 D	2.58	482	220	753	343	2.43×10^{-5}	2.07×10^{-8}
10/90	3.39	580	229	689	274	2.54×10^{-5}	3.09×10^{-8}
25/75	3.07	585	184	691	334	2.85×10^{-5}	2.27×10^{-8}
50/50	4.49	330	174	691	284	7.79×10^{-5}	6.93×10^{-8}
0/100 T	1.2	807	237	1019	789	0.63×10^{-5}	0.18×10^{-8}
10/90	1.5	730	213	950	602	0.97×10^{-5}	0.36×10^{-8}
25/75	3.3	595	194	685	324	2.86×10^{-5}	2.50×10^{-8}
50/50	3.94	325	160	670	276	7.50×10^{-5}	6.55×10^{-8}

$*v_{\max}$ is the maximum rate of thermal decomposition; T_p is the temperature of the maximum rate of thermal decomposition; T_e is the temperature of the onset of thermal decomposition; T_f is the temperature of the complete thermal decomposition; $\Delta T_{1/2}$ is the temperature in the half-period of reaching the maximum rate of thermal decomposition; D_i is the index of the onset of thermal decomposition of fuels; and D_f is the index of the complete thermal decomposition of fuels.



In this case, salts (sulfates), which are the components of ash, water vapor (which can participate in the subsequent carbon oxidation reactions with the formation of hydrogen), and hydrogen were formed. In turn, hydrogen can interact with sulfur oxides to form sulfuric acid, etc.

In recent years, foreign researchers have established the possibility of the occurrence (at temperatures over 900°C) of the reactions



which lead to the sequestration of sulfur and nitrogen oxides on their interactions with hydrogen or oxygen.

The regularities established in the above experiments illustrate the synergistic effect of the combustion of mixed fuels and the need for a detailed study of not only coal, which is the basis of these fuels, but also the mixed fuels at the stage of their formation. In this

case, a maximum positive effect of the use of fuels based on coal and wood can be achieved.

CONCLUSIONS

As a result of the experimental studies, we established that, in the joint thermal decomposition of lean coal and wood particles, the process of coal pyrolysis is shifted to lower temperatures. Furthermore, the emission of sulfur and nitrogen oxides in the joint pyrolysis of coal and wood decreases significantly (to 90%), as compared with homogeneous coal. We formulated a hypothesis on the mechanism of oxide sequestration in the joint thermal decomposition of coal and wood because of the thermochemical interaction of intermediate pyrolysis products (sulfur oxides, water vapor, and metal oxides) with the formation of metal salts in solid products. We found that the effect also manifested itself in the pyrolysis of a mixture of long-flame coal and wood, but in a significantly smaller volume. The results obtained demonstrate that it is reasonable to experimentally study the processes of the joint thermal decomposition of dispersed coal and shredded wood at the stage of the formulation of composite fuels. The results of the experiments are the basis for the development of mathematical models of the combustion of a mixture of coal and wood biomass in terms of the description of thermal processes occurring at the stage of the thermal decomposition of a mixture of these two fuels.

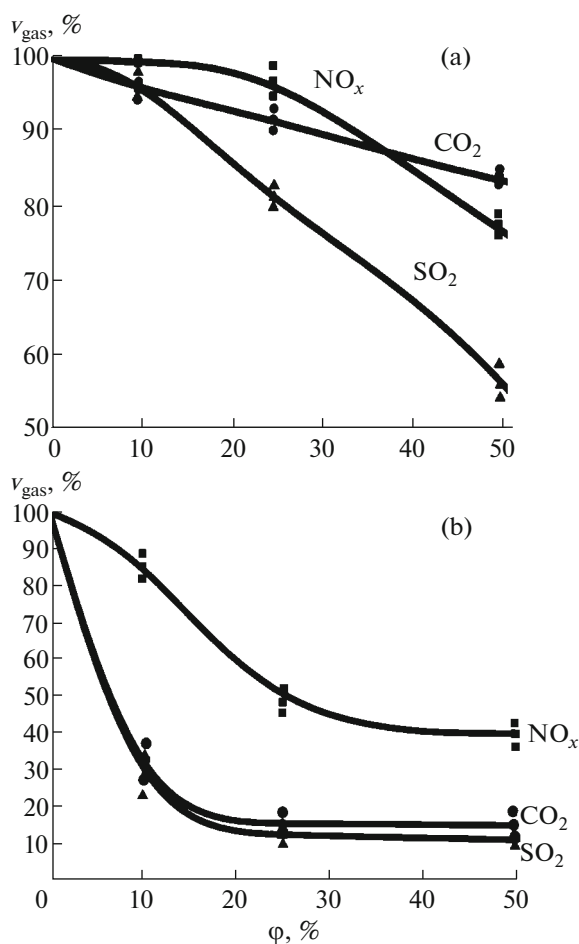


Fig. 4. Concentrations of the main harmful emissions upon the thermal decomposition of composite fuels based on coals of grades (a) D and (b) T; ϕ is the fraction of wood in the fuel.

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